

Análise PESTEL

Introduction

This PESTEL analysis was developed to evaluate the external macro-environmental factors that may directly or indirectly influence the development, adoption, scalability, and strategic positioning of Bayer's agricultural decision-support simulator MVP. Rather than analyzing the project exclusively from a technical or operational perspective, this framework broadens the assessment by examining how Political, Economic, Social, Technological, Environmental, and Legal dimensions shape both the viability of the solution and its long-term relevance within Brazilian agribusiness.

Given that the proposed simulator operates in a highly complex agricultural ecosystem, one that depends on climate conditions, public agricultural policy, producer profitability, digital infrastructure, social trust, environmental sensitivity, and regulatory compliance, its success cannot be determined solely by algorithmic performance. The project exists within a broader system where variables such as rural credit, pesticide regulation, precision agriculture trends, climate volatility, ESG pressures, and legal accountability all influence whether the tool will be adopted, trusted, and strategically valuable.

This analysis therefore seeks to identify not only the opportunities that may strengthen the project, but also the structural risks, dependencies, and external constraints that could impact implementation. By understanding these six dimensions in an integrated way, the project team can better anticipate barriers, prioritize strategic decisions, and align the MVP with both Bayer's operational objectives and the real conditions faced by growers, agronomists, and the agricultural market.

Ultimately, this PESTEL framework serves as a strategic decision-making foundation, allowing the project to move beyond a purely technological solution and toward a more resilient, context-aware, and scalable product capable of operating effectively in the real-world dynamics of modern agribusiness.

PPOLITICAL

Description

In this analysis, the Political factor needs to be understood less as "partisan politics" and more as public policy, state risk management, trade regulation, and institutional data infrastructure. From this point of view, the most relevant political factor for the project is ZARC (Agricultural Climate Risk Zoning) together with MAPA (Ministry of Agriculture and Livestock), used as a technical tool by the federal government that indicates the best planting periods for several crops in Brazil, aiming to reduce losses caused by adverse climate. (BRAZIL)

Along with this, ZARC connects to two important programs of the Brazilian government, namely: Proagro and PSR. Proagro works as a protection for the producer who financed the planting, covering part of the losses when there are problems such as drought, excessive rainfall, pests, or diseases. PSR, on the other hand, helps make rural insurance cheaper, because the government pays part of the value of the policy contracted by the producer. In the 2025–2027 plan, the government foresees subsidies of up to R\$ 60 thousand per group of activities and up to R\$ 120 thousand per producer per year, which demonstrates that these mechanisms have real weight in the sector's risk management. (BRAZIL)

In the geopolitical sphere, in addition to Brazilian agricultural policies, global political factors that directly affect agribusiness must also be considered. The recent escalation of conflicts on the international stage, especially in the Middle East, has increased pressure on energy and fertilizer markets, raising costs and expanding uncertainty in global agrifood chains. This factor is especially relevant for the Brazilian context, since the country still imports around 85% of the fertilizers it uses, which increases its exposure to external shocks. In addition, crops such as soybeans and corn are among the ones that depend most on these inputs, making them more sensitive to geopolitical and supply fluctuations. (BRAZIL)

Finally, there is also a quieter but real political dependency: the partner's TAPI relies precisely on public databases and government or international APIs. Whenever a project starts to depend on public data on climate, zoning, and remote sensing, it also starts to derive from the governance, standardization, and institutional reliability of those sources.

Impact on the Project

An agronomically correct recommendation can lose strength if it is not aligned with the institutional and financial environment in which the producer actually makes decisions. Likewise, geopolitical instability does not change the central agronomic logic of the simulator, but it directly influences the client's decision environment. In scenarios of rising input prices and greater risk perception, the producer may prioritize recommendations focused on operational safety, margin preservation, and return on investment, instead of seeking exclusively maximum productivity.

Why it Matters

This factor matters because the simulator will not operate in an isolated technical environment. It will be inserted into a real agricultural system shaped by public policy, rural insurance mechanisms, government programs, and international political instability. Therefore, political conditions influence not only the feasibility of adoption, but also the kind of recommendation that will be considered valuable by the producer and by Bayer's field teams.

Monitoring Consideration

This factor should be monitored through updates in Brazilian agricultural public policy, especially ZARC, MAPA, Proagro, and PSR rules, as well as changes in the availability and reliability of public databases used by the project. In addition, international geopolitical developments that affect energy and fertilizer prices should also be monitored, since they can shift the producer's decision criteria and risk tolerance.

ECONOMIC FACTORS

1. Volatility of Agricultural Commodity Prices

Description

Agricultural commodity prices, especially soybeans and corn, are strongly influenced by global supply and demand, climate events, exchange-rate variation, and geopolitical instability. According to the FAO Food Price Index, the global index averaged 127.2 points in 2025, which was 4.3% above 2024 levels, showing that food markets remain unstable rather than fully stabilized. In Brazil, CEPEA data showed that soybean prices in 2025 reached their lowest real annual average since 2019, even during strong harvest volumes. This means growers may experience major profitability swings even in productive years. Because Bayer's simulator targets producers whose investment behavior depends directly on profitability and economic confidence, these fluctuations can significantly affect willingness to adopt new technologies.

Impact on the Project

This volatility directly affects growers' willingness to invest in innovation. During stronger pricing cycles, producers are generally more willing to adopt digital tools that improve operational efficiency and long-term planning. During weaker or uncertain cycles, many growers prioritize short-term financial security and may delay investments unless the technology clearly reduces immediate costs.

Why it Matters

This matters because commodity price volatility affects more than grower income, it can directly shape Bayer's adoption environment. If soybean or corn prices remain strong, Bayer's simulator may be positioned mainly as a strategic optimization tool. If prices fall significantly, growers may become more cautious, meaning Bayer may need to emphasize immediate cost control, waste reduction, and margin protection instead. This makes commodity volatility relevant not only economically, but also strategically, because it can influence adoption speed, customer urgency, and how Bayer communicates product value.

Monitoring Consideration

Commodity price trends should be monitored throughout future sprints because continued volatility may significantly affect grower confidence, urgency, and adoption behavior.

2. Rising Agricultural Input Costs

Description

The rising cost of fertilizers, fuel, pesticides, and crop protection products has become one of the strongest economic pressures affecting agriculture. According to the World Bank's fertilizer price monitoring, fertilizer prices rose sharply during recent years due to global supply disruptions and geopolitical instability, while FAO market analyses also highlight increased pressure on agricultural production costs. As production expenses rise,

inefficiency becomes more expensive, and operational mistakes such as poor spraying timing or wasteful input allocation can have larger financial consequences.

Impact on the Project

As production costs increase, growers face greater pressure to optimize decisions related to planting, spraying, and resource use. This directly strengthens the practical relevance of Bayer's simulator because precision becomes more financially valuable when every operational inefficiency costs more.

Why it Matters

This factor is important because rising input costs can increase demand for Bayer's simulator rather than reduce it. As fertilizers, fuel, and crop protection products become more expensive, producers may place greater value on systems that reduce waste and improve efficiency. This means Bayer's simulator may shift from being seen mainly as a strategic advantage to being viewed as a more necessary operational tool for protecting margins.

Monitoring Consideration

Input cost indicators should be monitored throughout future sprints because sustained inflation may significantly increase urgency for efficiency-focused technologies.

3. Interest Rates and Access to Agricultural Credit in Brazil

Description

In Brazil, many growers depend on financing and rural credit to modernize operations and adopt innovation. Banco Central do Brasil data shows that interest rates and agricultural credit conditions strongly influence investment flexibility. Because many producers rely on borrowed capital for operational and strategic upgrades, higher borrowing costs can reduce their ability or willingness to invest in new technologies.

Impact on the Project

Even if Bayer's simulator demonstrates strong technical value, adoption may slow if financing conditions become restrictive. High interest rates may push growers to prioritize essential operational expenses over digital transformation.

Why it Matters

This matters because financing conditions can create a structural adoption barrier independent of Bayer's product quality. Even if the simulator provides strong strategic value, restrictive credit environments may reduce implementation speed. This means Bayer may need to emphasize affordability, practical ROI, and accessible implementation in financially constrained periods.

Monitoring Consideration

Brazilian interest rates, Selic cycles, and rural credit conditions should be monitored throughout future sprints because financing shifts may directly influence adoption potential.

SOCIAL

1. Aging and Digital Inequality in Rural Areas

Description

The simulator's target audience is aging: 23.2% of Brazilian rural producers are over 65 years old, while only 2% are under 25 (IBGE, 2019). At the same time, 65.47% of family farming operations have an education level of up to primary school (IBGE, 2019), and 4G/5G coverage in rural areas, though growing, reached only 33.9% of rural areas in 2025 (ConectarAgro, 2025).

Impact on the Project

A complex interface or one dependent on stable connectivity excludes a large portion of the audience the project intends to reach. The risk is that the simulator becomes effective only for an already-technified segment of producers — which limits scale, reduces social impact, and concentrates the benefit among those who already hold a competitive advantage.

Why It Matters

Digital inclusion in the field is a prerequisite for the project to fulfill its potential contribution to SDGs 8 and 10 (prosperity and reduced inequalities) (UN, 2015). A product that deepens the technological gap between large and small producers contradicts Bayer's ESG narrative and may attract institutional criticism.

2. Pesticide Drift and Public Health

Description

In 2025, Brazil recorded 9,729 cases of unintentional pesticide poisoning (Brasil, 2026), the highest number since 2015. Technical studies indicate that only 32% of the applied product reaches its target, with approximately 19% dispersing into the air as drift (Embrapa, 2020-2024) — affecting rural workers, neighboring communities, and watersheds.

Impact on the Project

The simulator operates at exactly this critical point: optimizing the application window has direct potential to reduce drift and, consequently, poisoning cases. If communicated with verifiable data, this represents a robust ESG asset. If poorly communicated — or if a simulator recommendation becomes associated with a drift incident — the reputational damage could be severe and difficult to reverse.

Why It Matters

Public health associated with pesticide use is the main vector of social pressure on companies in this sector. Organizations such as ABRASCO and Repórter Brasil actively monitor this topic (Repórter Brasil, 2026), and any incident associated with the product can gain disproportionate national attention relative to the size of the project.

3. Resistance from Intermediaries: Agronomists and Cooperatives

Description

Agronomists and Technical Advisors (RTVs) are the primary trust channels for rural producers — and WhatsApp is the number one transaction channel in Brazilian agribusiness

(McKinsey, 2024). Cooperatives such as Coamo, Cocamar, and CCAB provide direct technical assistance to their members and may perceive the direct Bayer–producer channel as a threat to their intermediation role.

Impact on the Project

If agronomists and cooperatives feel their technical authority is being threatened by the simulator, they can act as adoption barriers rather than accelerators. On the other hand, if engaged as co-pilots and product validators, they become the main vector of scale and credibility.

Why It Matters

Trust in agribusiness is built locally through human relationships (McKinsey, 2024). A digital product that ignores this relational logic will struggle to scale — regardless of its technical quality. The proactive engagement of these actors is not optional; it is a condition of commercial viability.

Monitoring Consideration

Social dimension monitoring should track all three identified factors simultaneously. Regarding the producer profile and digital inclusion (1.), the team should record at each user testing round whether producers of different age groups and connectivity levels are able to use the simulator without external support — usability difficulties are design alerts, not merely communication issues. Regarding pesticide drift and public health (2.), it is recommended to actively monitor media coverage of the pesticide topic in Brazil throughout the project: if the subject gains negative traction, the simulator's communication plan should be proactively reviewed, and any drift incident occurring during the period should be documented and analyzed for potential association with the product. Regarding resistance from agronomists and cooperatives (3.), acceptance levels should be formally recorded during validation sessions — responses indicating discomfort with the simulator's role or a perception of disintermediation should be treated as priority inputs for product adjustment before the next iteration.

TECHNOLOGICAL FACTORS

1. Expansion of Data Integration via APIs and External Data Sources

Description

Bayer's simulator depends heavily on the integration of multiple internal and external datasets, including weather forecasts, meteorological history, satellite imagery, vegetation indices, and commodity databases. Public systems such as NASA Earthdata, INMET, Sentinel, and Landsat significantly expand available agricultural intelligence. The growing availability of these systems creates major opportunities for better predictive capability, but also increases dependence on API reliability, synchronization quality, and data consistency.

Impact on the Project

More available data can strengthen simulator sophistication by improving predictive recommendations and contextual precision. However, Bayer's system also becomes increasingly dependent on stable external integrations. API failures, inconsistent updates, or unreliable synchronization can directly reduce recommendation quality.

Why it Matters

This matters because Bayer's simulator is fundamentally dependent on transforming multiple fragmented data sources into reliable operational intelligence. If external data systems are unstable or poorly integrated, Bayer's recommendation quality may decline, which can damage user trust and platform credibility. This makes data integration not just a technical advantage, but a strategic dependency.

Monitoring Consideration

API reliability, external data stability, and integration performance should remain high-priority monitoring areas throughout future sprints.

2. Growth of Precision Agriculture and Data-Driven Farming

Description

Precision agriculture is expanding rapidly through IoT, predictive analytics, remote sensing, and data-driven operational systems. According to the World Bank and McKinsey, digital agriculture adoption continues to grow because producers increasingly seek productivity, sustainability, and operational efficiency improvements. This reflects a broader industry movement toward integrated digital decision-making.

Impact on the Project

This trend supports Bayer's strategic direction because growers are increasingly expecting systems that move beyond simple dashboards toward recommendation-oriented platforms that actively improve decisions.

Why it Matters

This matters because the growth of precision agriculture strengthens Bayer's market relevance while also increasing competitive expectations. Bayer's simulator is aligned with a growing technological shift, but as adoption increases, growers may also expect greater sophistication, trust, and performance. This means Bayer must remain competitive not just by being digital, but by delivering strong practical value.

Monitoring Consideration

Precision agriculture adoption rates and digital farming trends should be monitored continuously to ensure Bayer's simulator remains aligned with market expectations.

3. Data Quality, Reliability, and Interoperability Challenges

Description

Although digital agriculture is expanding, data fragmentation, poor interoperability, inconsistent standards, and incomplete datasets remain major barriers. CGIAR and World Bank digital development analyses both highlight that data volume alone does not guarantee value unless systems can properly validate, standardize, and integrate information.

Impact on the Project

For Bayer's simulator, poor data quality can create direct strategic risk. Inaccurate or fragmented data may lead to flawed recommendations, reducing both technical effectiveness and user trust.

Why it Matters

This matters because Bayer's simulator depends not only on advanced analytics, but on reliable and trustworthy outputs. If recommendation quality is compromised by poor data governance, Bayer may face credibility challenges that affect long-term adoption. This makes data quality a foundational issue for scalability, trust, and product legitimacy.

Monitoring Consideration

Data governance, validation systems, and interoperability performance should remain strategic priorities throughout development.

ENVIRONMENTAL

Description

In the context of this project, the Environmental factor is one of the most determining, because it directly affects the main use case of the MVP. The documentation itself (the Partner's TAPI) defines that the spraying recommendation depends on the combination of climatic factors such as temperature, wind, humidity, time of day, risk of thermal inversion, and short-term rain. In line with this, Embrapa highlights that wind, temperature, and relative humidity act directly on spraying, and that temperatures above 30 °C and humidity below 55% favor droplet evaporation and increase the risk of efficiency loss and drift (EMBRAPA).

Furthermore, there is also the factor of climate volatility in the global scenario lately. The WMO reported that Latin America in 2024 had an increase in average temperature of around 0.90 °C above the 1991–2020 average and highlighted the impacts of heat waves, droughts, and intense rainfall on agriculture and food security. On the other hand, Conab showed, in the crop monitoring of April 2026, that rainfall distribution supported a positive scenario for second-crop corn in Mato Grosso, while irregular rainfall and high temperatures reduced the productive potential in other states (FAO).

In the planting and plantability part, the environmental role remains evident. Embrapa indicates that the ideal corn population varies according to water availability, soil fertility, sowing season, and spacing; in parallel, soybeans have a greater capacity for compensation, or "plasticity," than corn, as also confirmed by the partner in the meeting on 04/24/2026. Another study by Embrapa itself highlights, for second-crop corn in Mato Grosso, that productivity variation is strongly related to climatic conditions during critical periods. (PEREIRA)

Finally, there is still another environmental point in the strategic scenario of the project: the Bayer partner's presentation associates the platform with optimization of water use, localized reduction of pesticides, carbon footprint calculation, and regenerative agriculture.

Impact on the Project

If variables such as temperature, wind, humidity, rain, and thermal inversion risk are poorly measured, poorly predicted, poorly correlated, or poorly assessed, it will result in an MVP simulator with imprecise results. In addition, environmental instability can affect both negatively and positively the location and quality of the simulator's recommendation. In

product terms, this means that the simulator cannot issue a single recommendation for everyone; it needs to segment by crop, soil type, seasonality, and water risk, which originate from environmental factors. The environmental factor also affects not only the accuracy of the decision, but also the positioning of the product in relation to its clients, field teams, and sustainability programs.

Why it Matters

This factor matters because environmental conditions are not only background context — they are part of the core logic of the recommendation itself. The success of the simulator depends on its ability to interpret changing climate and field conditions correctly. In other words, the environmental factor is simultaneously an input, a quality criterion, and a business argument.

Monitoring Consideration

This factor should be continuously monitored through public and technical sources related to weather, climate, soil, and crop productivity. Since the environmental scenario is not predictable and not static, the project should pay close attention to changes in rainfall distribution, temperature variation, droughts, heat waves, and crop-specific environmental behavior, especially for soybeans and corn in key regions such as Mato Grosso.

LEGAL

1. Producer Data Protection (LGPD)

Description

Brazil has a fully enforced personal data protection law (Brasil, 2018). Location and productivity data from farms owned by individuals fall under its scope — and the agribusiness sector is one of the least prepared for this type of compliance in the country. The collection, storage, and sharing of this data require a defined legal basis, a clear privacy policy, and, when necessary, explicit consent from the producer.

Impact on the Project

Without a formal mapping of what data is collected, how it is used, and who is responsible for it — Bayer or Inteli — the project is exposed to administrative sanctions and loss of producer trust. The risk is compounded by the fact that agricultural data has high commercial and strategic value, which increases the interest of third parties and the sensitivity of data subjects.

Why It Matters

Compliance with data protection legislation is not just a legal obligation — it is a condition of trust. Producers who do not trust how their data is used simply will not adopt the tool, regardless of its technical quality. Furthermore, public data such as INMET and Sentinel/Landsat satellites are freely available (IBGE, 2019), which represents a strategic advantage: part of the simulator's database carries no intellectual property risk whatsoever.

2. Agronomic Prescription Requirements and Civil Liability

Description

Brazilian law requires that any decision to apply pesticides be signed off by a licensed agronomist (Brasil, Pesticides Act, 2023). No software can legally replace this professional. In addition, the Consumer Protection Code establishes strict liability for service defects (Brasil, 1990) — meaning that if the simulator recommends an inadequate application window that causes harm to the producer, workers, or third parties, both Bayer and Inteli can be held liable regardless of intent.

Impact on the Project

The simulator must be designed from the outset as a decision-support tool for the agronomist — never as a substitute. This is not just a positioning choice: it is a legal requirement. Visible warnings in the interface, auditable logs of every recommendation generated, and disclaimers in the terms of use are non-negotiable measures to reduce the project's legal exposure.

Why It Matters

A single documented incident of drift associated with a simulator recommendation could trigger lawsuits, negative media coverage, and regulatory review — even if the product is technically sound. Regulatory Standard NR-31 already explicitly prohibits application in meteorological conditions with a risk of drift (Brasil, 2022), which means the simulator must, at minimum, alert users when conditions fall outside the safe limits recommended by product labels.

3. AI Legal Framework and Governance of Automated Systems

Description

Brazil is advancing the approval of a specific law for artificial intelligence systems — passed by the Senate in December 2024 and under review in the Lower House in 2026 (Brasil, AI Legal Framework, 2023). Although the simulator will likely be classified as a low-risk system, the legislation already demands transparency, traceability of automated decisions, and human oversight. The international standard ISO/IEC 42001:2023 provides a voluntary AI management framework aligned with these principles (ISO/IEC, 2023).

Impact on the Project

Even as an academic prototype, the simulator must be designed with auditable logs of generated recommendations, mechanisms that ensure the producer understands how the recommendation was built, and the ability to question or ignore it. This is not just a future requirement — it is a good practice that legally protects the project and facilitates Bayer's ESG audits globally.

Why It Matters

The AI regulatory landscape is evolving rapidly. Projects that do not incorporate governance principles from the start will require costly rework when obligations become formal. Voluntary adoption of ISO/IEC 42001:2023 (ISO/IEC, 2023) positions the project ahead of the regulatory curve and generates differentiation in Bayer's sustainability audits with European investors and partners.

Monitoring Consideration

Legal dimension monitoring should ensure that all three identified factors are continuously covered throughout the project sprints. Regarding data protection (1.), the team should verify at each sprint whether new types of data have begun to be collected or processed without adequate contractual and legal coverage — any expansion of the data scope requires an immediate review of the privacy mapping and confirmation that the responsibility roles between Bayer and Inteli remain formally defined. Regarding civil liability and agronomic prescription requirements (2.), each new version of the simulator should be reviewed for the visibility of advisory disclaimers in the interface and the correct functioning of the recommendation log, with particular attention to edge-case scenarios where the system could suggest conditions close to the limits prohibited by Regulatory Standard NR-31 (Brasil, 2022). Regarding the AI Legal Framework (3.), the team should monitor the progress of Bill 2338/2023 in the Chamber of Deputies: any approval, amendment, or new governance requirement should be assessed for its impact on the simulator's requirements while the project is still ongoing, avoiding rework at the final delivery stage.

Conclusion/Ponderations:

With the PESTEL analysis completed in a full way and compatible with the context of the project with Bayer, it is feasible to carry out a weighting to identify which of the variables within the PESTEL analysis has the greatest impact on the final MVP:

1. Environmental

This is the factor with the greatest weight because it is directly linked to the problem and the central logic of the MVP. This is explainable because variables such as temperature, wind, humidity, rain, soil, and water risk are not just context, but the main inputs of the MVP. As these variables vary according to crop, region, and season, it can be concluded that the environmental factor has the greatest direct impact on the quality of the recommendation.

2. Technological

Technology appears right after because the project depends on integration, information provided by the partner, and the transformation of this data into reliable decision support. Examples such as APIs, machine data, satellite images, and public databases only add value if they are well connected, explained, standardized, and reliable. Therefore, technology is not just support, but one of the foundations of the simulator.

3. Economic

The economic factor is relevant because the adoption of the tool depends on the financial reality of the producer. Since our simulator will be based on the client's profile, the variation in commodity prices, the increase in input costs, and access to credit influence whether the simulator will be seen as a strategic optimization tool or as a necessary tool to protect margin and improve ROI.

4. Political

The political factor is relevant because the simulator is inserted into a real

agricultural system, in which it is constantly influenced by public policies, risk management programs, and geopolitical pressures. Elements such as ZARC, Proagro, PSR, and fertilizer dependency affect the way the recommendation will be evaluated in practice. Furthermore, global crises such as those in the Middle East put pressure on fuel prices, which directly affects commodity costs.

5. **Legal**

The legal factor can define the limits of the solution's functioning, communication, and responsibility. Issues such as data protection, pesticide regulation, AI governance, and civil liability are important for the project's compliance and credibility. Even more, since this is an academic project in which data leakage can occur accidentally, but at the same time these are not the main technical value risks of the simulator.

6. **Social**

Finally, the least relevant but not negligible factor is the social factor. It mainly affects the adoption, communication, and reputation of the tool. Acceptance by producers, agronomists, and cooperatives, in addition to issues of digital inclusion and perception about pesticides, can influence the scale and legitimacy of the project. Even so, it has a more indirect impact on the central functioning of the simulator than the other factors.

References:

AMERICAN FARM BUREAU FEDERATION. Privacy and Security Principles for Farm Data. Washington, D.C., 2024.

BANCO CENTRAL DO BRASIL. Crédito rural. Brasília: Banco Central do Brasil, 2026. Available at: <https://www.bcb.gov.br/estatisticas/creditorural>. Accessed: Apr. 28, 2026.

BANCO CENTRAL DO BRASIL. Taxas de juros básicas (Selic). Brasília: Banco Central do Brasil, 2026. Available at: <https://www.bcb.gov.br/en/monetarypolicy/interestrates>. Accessed: Apr. 28, 2026.

BAYER. Environmental and Climate Protection. Bayer Brasil, 2024. Available at: <https://www.bayer.com.br/pt/sustentabilidade/protecao-ambiental-e-climatica>. Accessed: Apr. 2026.

BRASIL. Agência Nacional de Proteção de Dados (ANPD). Resolução CD/ANPD nº 4, de 24 de fevereiro de 2023. Regulation on Dosimetry and Application of Administrative Sanctions. Brasília: ANPD, 2023. Available at: <https://www.gov.br/anpd>. Accessed: Apr. 2026.

BRASIL. Lei nº 8.078, de 11 de setembro de 1990. Consumer Protection Code. Brasília: Presidência da República, 1990. Available at: https://www.planalto.gov.br/ccivil_03/leis/l8078compilado.htm. Accessed: Apr. 2026.

BRASIL. Lei nº 13.243, de 11 de janeiro de 2016. Science, Technology and Innovation Legal Framework. Brasília: Presidência da República, 2016. Available at: https://www.planalto.gov.br/ccivil_03/_ato2015-2018/2016/lei/l13243.htm. Accessed: Apr. 2026.

BRASIL. Lei nº 13.709, de 14 de agosto de 2018. General Personal Data Protection Law (LGPD). Brasília: Presidência da República, 2018. Available at: https://www.planalto.gov.br/ccivil_03/_ato2015-2018/2018/lei/l13709.htm. Accessed: Apr. 2026.

BRASIL. Lei nº 14.785, de 27 de dezembro de 2023. Pesticides, Environmental Control Products and Related Items Act. Brasília: Presidência da República, 2023. Available at: https://www.planalto.gov.br/ccivil_03/_ato2023-2026/2023/lei/l14785.htm. Accessed: Apr. 2026.

BRASIL. Ministério da Saúde. Sistema de Informação de Agravos de Notificação (SINAN). Unintentional pesticide poisoning cases, 2015–2025. Brasília: Ministério da Saúde, 2026.

BRASIL. Ministério do Trabalho e Previdência. Portaria MTP nº 4.219, de 20 de dezembro de 2022. Regulatory Standard nº 31 — Health and Safety in Agricultural Work. Brasília, 2022.

BRASIL. Projeto de Lei nº 2.338, de 2023. Artificial Intelligence Legal Framework. Passed by the Federal Senate in December 2024; under review in the Chamber of Deputies as of April 2026. Brasília: Senado Federal, 2023.

CENTRO DE ESTUDOS AVANÇADOS EM ECONOMIA APLICADA (CEPEA). Indicador da soja. Piracicaba: CEPEA/ESALQ-USP, 2026. Available at: <https://www.cepea.esalq.usp.br/br/indicador/soja.aspx>. Accessed: Apr. 28, 2026.

CGIAR. Interoperability in Digital Agriculture Systems. Montpellier: CGIAR, 2026. Available at: <https://cgspace.cgiar.org/handle/10568/114112>. Accessed: Apr. 28, 2026.

CONECTARAGRO. Rural Connectivity Index (ICR) 2025. São Paulo: ConectarAgro, 2025.

CONSELHO NACIONAL DE AUTORREGULAMENTAÇÃO PUBLICITÁRIA (CONAR). Brazilian Advertising Self-Regulatory Code — Update of Articles 36-A and 36-B. São Paulo: CONAR, Oct. 2025.

EMBRAPA. Studies on Pesticide Application Efficiency and Drift. Brasília: Empresa Brasileira de Pesquisa Agropecuária, 2020–2024.

FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS (FAO). FAO Food Price Index. Rome: FAO, 2026. Available at: <https://www.fao.org/worldfoodsituation/foodpricesindex/en/>. Accessed: Apr. 28, 2026.

FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS (FAO). Markets and Trade. Rome: FAO, 2026. Available at: <https://www.fao.org/markets-and-trade/en/>. Accessed: Apr. 28, 2026.

IBGE. Agricultural Census 2017. Rio de Janeiro: Instituto Brasileiro de Geografia e Estatística, 2019. Available at: <https://www.ibge.gov.br/estatisticas/economicas/agricultura-e-pecuaria/21814-2017-censo-agropecuaria.html>. Accessed: Apr. 2026.

INSTITUTO NACIONAL DE METEOROLOGIA (INMET). Portal INMET. Brasília: INMET, 2026. Available at: <https://portal.inmet.gov.br/>. Accessed: Apr. 28, 2026.

ISO/IEC 42001:2023. Information Technology — Artificial Intelligence — Management System. Geneva: International Organization for Standardization, 2023.

MCKINSEY & COMPANY. Agriculture Technology Insights. New York: McKinsey & Company, 2026. Available at: <https://www.mckinsey.com/industries/agriculture/our-insights/agriculture-technology>. Accessed: Apr. 28, 2026.

MCKINSEY & COMPANY. The Digital Future of Agribusiness: Brazil Leads the Way. New York: McKinsey & Company, 2024.

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION (NASA). Earthdata. Washington, D.C.: NASA, 2026. Available at: <https://earthdata.nasa.gov/>. Accessed: Apr. 28, 2026.

PEREIRA, André Ferreira et al. Evaluation of plant populations of second-crop corn hybrids in the region of Ipameri, southeast of Goiás, in 2013. Santo Antônio de Goiás: Embrapa, 2013. Available at: <https://www.alice.cnptia.embrapa.br/alice/bitstream/doc/976553/1/33802.pdf>. Accessed: Apr. 29, 2026.

REPÓRTER BRASIL. Brazil Breaks Pesticide Poisoning Record, with 27 Cases per Day in 2025. São Paulo: Repórter Brasil, Mar. 2026. Available at: <https://reporterbrasil.org.br/2026/03/brasil-recorde-intoxicacao-agrotoxico/>. Accessed: Apr. 2026.

SINDIVEG. Drift: How Proper Droplet Size Selection Reduces Losses and Increases Application Safety. São Paulo: Sindicato Nacional da Indústria de Produtos para Defesa Vegetal, 2023. Available at: <https://sindiveg.org.br/artigos/deriva-como-a-selecao-adequada-do-tamanho-de-gotas-reduz-perdas-e-aumenta-a-seguranca-da-aplicacao/>. Accessed: Apr. 2026.

UNITED NATIONS (UN). Transforming Our World: The 2030 Agenda for Sustainable Development. New York: UN, 2015. Available at: <https://brasil.un.org/pt-br/91863-agenda-2030-para-o-desenvolvimento-sustentavel>. Accessed: Apr. 2026.

WORLD BANK. Digital Agriculture. Washington, D.C.: World Bank, 2026. Available at: <https://www.worldbank.org/en/topic/agriculture/brief/digital-agriculture>. Accessed: Apr. 28, 2026.

WORLD BANK. Digital Development. Washington, D.C.: World Bank, 2026. Available at: <https://www.worldbank.org/en/topic/digitaldevelopment>. Accessed: Apr. 28, 2026.

WORLD BANK. Fertilizer Prices. Washington, D.C.: World Bank, 2026. Available at: <https://www.worldbank.org/en/topic/agriculture/brief/fertilizer-prices>. Accessed: Apr. 28, 2026.